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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B. Tech (ECE)

Name of the Paper: Optical Fiber Communication

Time: 3 Hours

Semester: VIIth

Paper Code: TEC 751

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	CO-01
(a)	Draw and explain the general block diagram of optical fiber communication.	
(b)	Describe the block diagram of optical fiber communication and its advantages and disadvantages.	
(c)	Discuss the effective refractive index, group delay, and mode delay.	
Q2	(20 marks)	CO-02
(a)	A long single-mode optical fiber has an attenuation of 0.4 dB km ⁻¹ when operating at a wavelength of 1.32 μ m. The fiber core diameter is 8 μ m and the laser source bandwidth is 800 MHz. Compare the threshold optical powers for stimulated Brillouin and Raman scattering within the fiber at the wavelength specified.	
(b)	Discuss and compare the waveguide and material dispersion in detail.	
(c)	Discuss absorption losses in optical fibers, comparing and contrasting the intrinsic and extrinsic absorption mechanisms.	
Q3	(20 marks)	CO-03
(a)	Describe with the aid of suitable diagrams the mechanism giving the emission of light from an LED.	
(b)	Discuss the spontaneous and stimulated emission in detail.	
(c)	Discuss the distributed feedback laser in detail.	
Q4	(20 marks)	CO-04
(a)	Discuss the noise source in optical fiber communication in detail.	
(b)	Describe the avalanche photodiode in detail. Also discuss the characteristics of photodetection.	
(c)	Discuss and compare the PN photodiode and PIN photodiode in detail.	
Q5	(20 marks)	CO-05
(a)	Discuss the advantages and characteristics of EDFA.	
(b)	Discuss optical Ethernet. Also discuss coherent and non-coherent detection in detail.	
(c)	Describe and discuss the optical receiver block diagram in detail.	

19/12/24
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